

Practice Quiz: Cognition / Behavioral Paradigms

(Research Methods)

Name: __ Date: __

Part A: Multiple Choice

1. The dissociation principle in task design means: A) All variables should be confounded B) Tasks should be designed so changes in one computational variable produce different behavioral predictions than changes in another C) Only one variable matters D) Variables cannot be separated
2. Eye fixations during active inference are directed toward: A) Random locations B) High-uncertainty, high-information locations — maximizing expected information gain through epistemic foraging C) Only familiar objects D) The brightest areas
3. Pupil dilation to unexpected events reflects: A) Light sensitivity B) Prediction error magnitude — larger surprises produce larger pupil responses via locus coeruleus activation C) Fatigue D) Eye strain
4. In probabilistic reversal learning, post-reversal perseveration indicates: A) High intelligence B) Low estimated volatility (ω) — the agent's prior is that the environment is stable, causing slow detection of contingency changes C) High learning rate D) Random behavior
5. The information sampling task's stopping rule under Active Inference is: A) Fixed number of samples B) Sample until $EFE(\text{commit}) < EFE(\text{sample})$ — stop when the expected free energy of committing is lower than continuing to gather information C) Always sample everything D) Never sample
6. Tonic pupil size correlates with: A) Stimulus brightness B) Environmental volatility and decision uncertainty — larger pupils during uncertain or volatile conditions C) Eye color D) Age

7. Hierarchical Gaussian Filter (HGF) models capture learning by: A) Fitting a single learning rate B) Estimating multiple levels of learning — belief updates at one level depend on the precision-weighted prediction errors from the level below, with volatility estimates at higher levels modulating lower-level learning rates C) Ignoring hierarchy D) Using only point estimates

8. Reaction time under Active Inference reflects: A) Only motor speed B) Decision confidence — influenced by the precision of posterior beliefs and the expected free energy difference between policies, with faster RTs for higher-precision, more decisive inferences C) Only stimulus difficulty D) Only age

Part B: Short Answer

1. A participant in a probabilistic reversal learning task shows slow updating after reversals (high perseveration). Under an Active Inference model, this could be due to either low volatility estimates OR high prior precision. Describe how you would dissociate these two explanations using additional behavioral or model-fitting evidence. (200 words)

2. Explain why eye-tracking is particularly informative for testing Active Inference predictions about cognition. What specific predictions does Active Inference make about where people look and for how long, and how do these differ from salience-based models? (200 words)

Part C: Essay Questions

1. Design a comprehensive behavioral study testing Active Inference's predictions about the exploration-exploitation trade-off. Include: (a) two tasks that dissociate epistemic from pragmatic value, (b) manipulations of uncertainty and volatility, (c) measures (choices, RT, eye-tracking, pupillometry), (d) computational model specification, (e) parameter recovery validation, (f) expected results. (600 words)

2. Compare behavioral evidence for Active Inference with behavioral evidence for Reinforcement Learning on the same set of tasks (reversal learning, bandit tasks, planning). Where do the two frameworks make identical predictions? Where do they diverge? How would you design the critical experiment that distinguishes them? (400 words)

3. Write a methods section for a computational phenotyping study using behavioral data from a probabilistic learning task. Include: task description, model specification, fitting procedure, parameter recovery, group comparison analysis, and results reporting standards. (500 words)